

Anticancer Drugs

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OBJECTIVES:

- Describe the relevance of cell cycle kinetics to the mode of action and uses of different anticancer drugs
- Discuss the major classes of anticancer drugs : mechanisms of action , major toxicities and uses of the main drugs in each subclass
- Discuss the rationale underlying strategies of combination drug therapy

- **Cancer**: is a disease in which there is uncontrolled multiplication and spread within the body of abnormal forms of the body's own cells.
- **Differences between benign and malignant tumors**:
- **Malignant tumors** :Dedifferentiation, invasiveness, ability to metastasize (ability to spread to other sites through blood or lymphatics).
- Both types manifest uncontrolled proliferation.
- Differences from **normal cell**: uncontrolled proliferation, dedifferentiation, invasive and metastasize.

Approaches to treating established cancer

- Surgical excision of localized tumor (ideal).
- Irradiation.
- Chemotherapy (the role of each depends on the type of the tumor and its stage of development). Usually indicated for disseminated tumors and as supplement after surgical excision and irradiation.
- **The goal of therapy:** is to achieve cure by eradication of every cancer cell and if not possible; palliation is the goal.

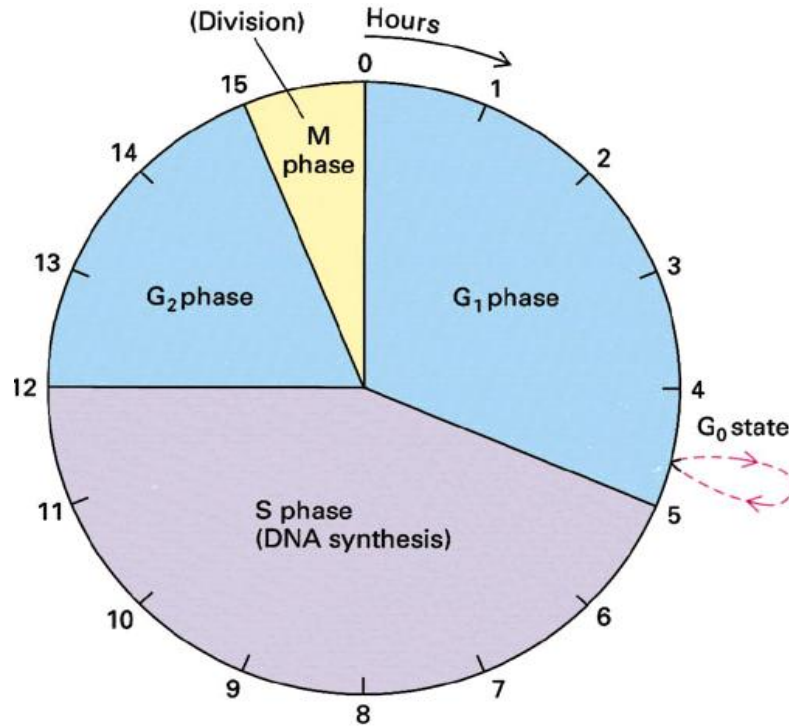
Cell cycle specificity of drugs:

- Generally, ***rapidly dividing tumors*** (e.g. hematologic tumors) are *more sensitive to anticancer drugs*
- While non-proliferating cells (**G₀ phase**) are more resistant.

Cont.,

- Chemotherapeutic agents that are **only effective against replicating cells** are called **cell cycle specific (CCS)** e.g. antimetabolites, bleomycin, vinca alkaloids and etoposide.
- while others are called **cell cycle non specific (CCNS)** and can affect **low-growth and high-growth fraction malignancies**, e.g. alkylating agents, cisplatin, nitrosurias,

Cell cycle



A

The cell cycle

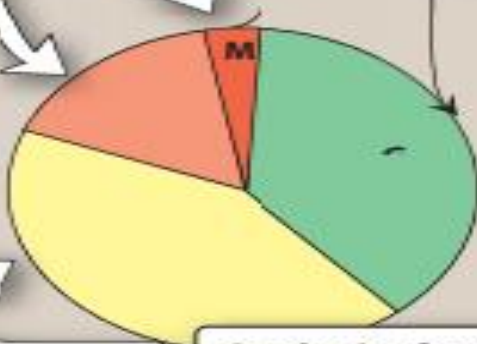
Synthesis of cellular components required for mitosis

Resting state
(cell is not dividing)

Mitotic phase
(cell divides)

DNA is replicated

Synthesis of enzymes needed for DNA synthesis



B

Cell-cycle specific drugs

Antimetabolites
Bleomycin peptide
antibiotics
Vinca alkaloids
Etoposide



Effective for high-growth-fraction malignancies, such as hematologic cancers

C

Cell-cycle non-specific drugs

Alkylating agents
Antibiotics
Cisplatin
Nitrosoureas



Effective for both low-growth-fraction malignancies, such as solid tumors, as well as high-growth-fraction malignancies

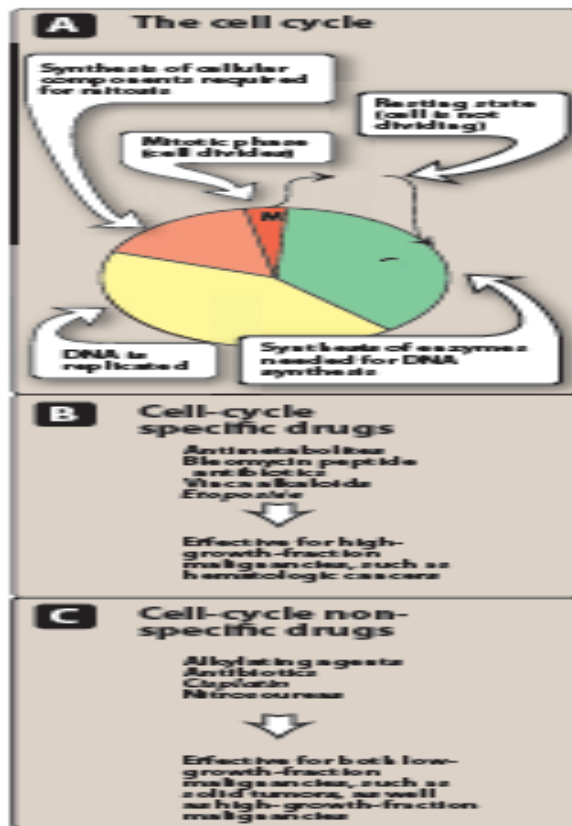


Figure 39.4
Effects of chemotherapeutic agents on the growth cycle of mammalian cells

Treatment regimens:

- Usually according to body surface area and is best made specific to each patient.
- - Chemotherapeutic agents follows **first order kinetics** which means **a constant fraction (percentage)** of cells destroyed by a given dose.
- NB : Some tumor cells are present in **sites** that can't gain access to by certain chemotherapeutic agents as CNS, thus intrathecal administration or irradiation may be used.

Combination therapy

*These combinations have **the advantages of:***

- *Maximal killing effect* with tolerated toxicity.
 - Affect *broader range* of cell lines.
 - Delay or *prevent resistance*.
-
- **Treatment protocols:** usually used intermittently and named by an acronym, e.g. POMP.

Problems associated with chemotherapy:

- **1-Development of resistance**
- **2-Multiple drug resistance (MDR).**
- **3-Toxicity:** cytotoxic drugs affect rapidly growing normal as well as cancer cells (e.g. cells of GIT mucosa, hair, bone marrow, buccal mucosa) →adverse effects.
- **4-Treatment-induced tumors:** As most of these agents are mutagens, tumors may arise many years (10 or more) after the cure of original tumor. This is specially a problem with alkylating agents.

Drugs used in cancer therapy:

- 1-Antimetabolites:

- synthetic analogues of normal metabolites, they interfere with purine or pyrimidine precursors either
They are CCS.

- **1-Folate antagonists:** Methotrexate (MTX)

- **2-Purine analogues:** 6-mercaptopurine

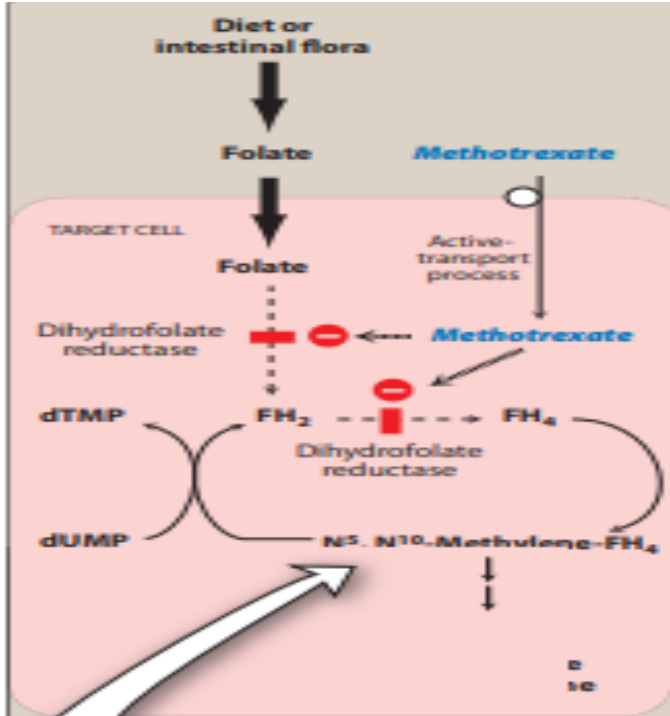
- **3-Pyrimidine analogues:** 5-Fluorouracil.

1-Folate antagonists: Methotrexate (MTX)

- Folates are essential for **DNA synthesis and cell division**. In order to act as coenzyme, folic acid must be converted to FH₄ (in 2 steps FH₂→FH₄) by **dehydrofolate reductase (DHFR)**.
- **MTX** becomes *polyglutamated*. Both **inhibit DHFR, inhibiting DNA, RNA and protein synthesis**→cell death.
- Administration of **leucovorin** or folinic acid (**leucovorin rescue**, when fatal dose of MTX is used to maximize effect; the patient will die if not given) to bypass the blocked enzyme and replenish folate pool.

Mechanism of action of methotrexate





Leucovorin rescue

Administer N^5 -formyl- FH_4 (leucovorin or folinic acid), which is converted to $\text{N}^5, \text{N}^{10}$ -methylene- FH_4 and, therefore, bypasses the inhibited reductase.

- **Adverse effects:**

- 1-Commonly seen with most anticancer drugs as ***nausea, vomiting, diarrhea, stomatitis, myelodepression and alopecia***. Some of these adverse effects are **prevented or reversed** by low dose **leucovorin** administration.
- 2-**Renal damage**: with high dose MTX , prevented by....
- 3-**Hepatic** toxicity by long term use.
- 4-**Pulmonary** toxicity: reversible.
- 5-**Neurologic**: with intrathecal administration(seizures, encephalopathy,.....

- **Therapeutic uses:**

- acute lymphocytic leukemia, breast cancer,...
- low dose is used in treatment of rheumatoid arthritis and psoriasis.

- **Resistance:**

- Resistance is associated with increased production of dihydrofolate reductase.

- **Contraindicated** in pregnancy (abortifacient and mutagenic).

2-Purine analogues:

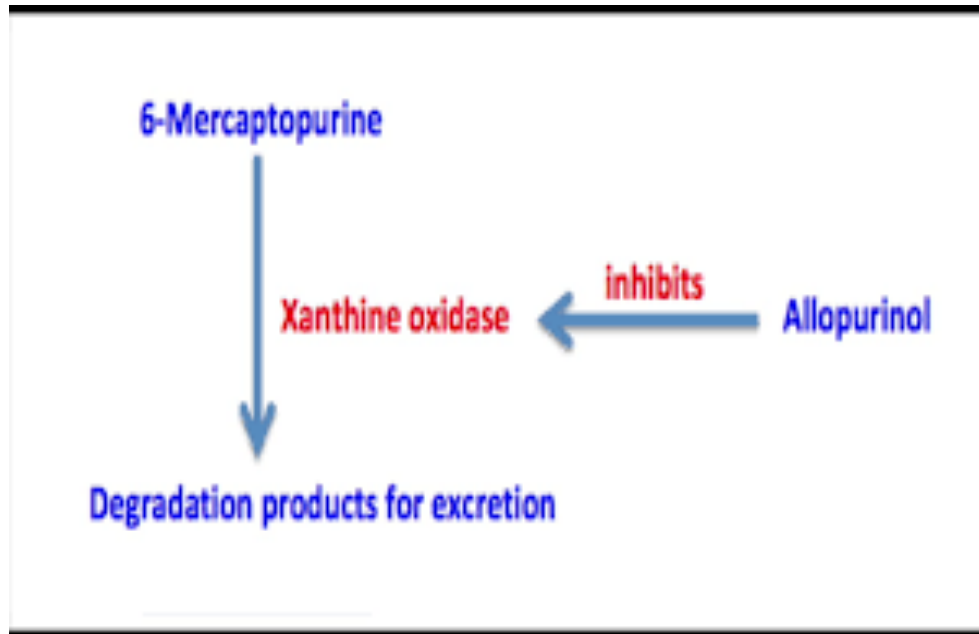
- 6-mercaptopurine (6-MP):

- It causes → non functional DNA and RNA .
- doesn't cross BBB.
- **Metabolized** to 6-MMP or thiouric acid by xanthine oxidase which is inhibited by **allopurinol** → accumulation of the drug and ↑ its toxicities So dose of 6-Mp should be decreased in patients taking allopurinol (by 25- 75%). **Salicylates, NSAIDs, sulphonamides and sulphonylureas enhance the toxicity of MXT.**

Adverse effects:

- Myelodepression, GIT disturbances and hepatotoxicity.

Interaction of 6-mercaptopurine with Allopurinol



6-Mercaptopurine



Allopurinol

Figure 39.11

Potential drug interaction between *allopurinol* and *6-mercaptopurine*.

3-Pyrimidine analogues:

- 5-Fluorouracil:

- Mechanism of action:

- Converted to FdUMP which competes with dUMP for thymidylate synthase.
- It forms a **complex** with the enzyme and its **coenzyme leucovorin** → ↓ thymidine & DNA synthesis and cell death.



Adverse reactions:

- GIT Upset and severe oral and mucosal ulceration, alopecia, myelodepression.
- **Allopurinol** mouth wash may be given to decrease oral toxicity.
- **Hand-foot syndrome** is seen after extended infusions (i.e. erythematous desquamation of hands and soles).
- **Uses:** slowly growing solid tumors.
- **N.B.:**
- **Leucovorin** is administered concomitantly with **5-FU** as the reduced folate coenzyme **is needed in the thymidylate synthase reaction**. Decreased coenzyme reduces the effectiveness of 5-FU.



II-Antibiotics:

- They produce their effects mainly by direct action on DNA causing disruption of DNA function. They are CCS.

- A -Bleomycin:

- Adverse effects :

- **pulmonary** toxicity is dose-limiting ranging from cough to fatal fibrosis.
- Hypertrophic **skin** changes and hyperpigmentation of hands are prevalent, while myelodepression is uncommon.



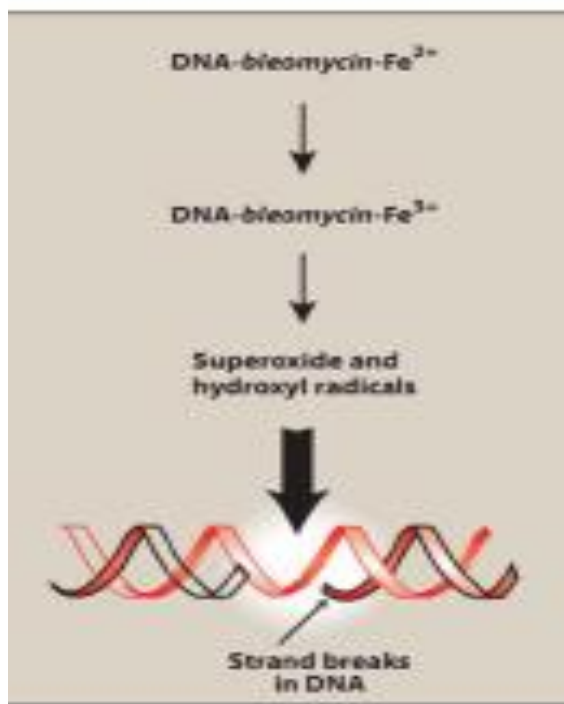


Figure 39.20

Bleomycin causes breaks in DNA by an oxidative process.

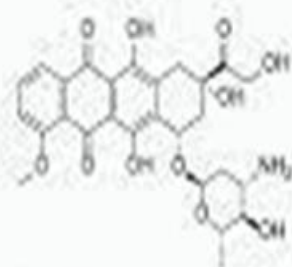
- **B -Doxorubicin and daunorubicin:**

- Doxorubicin is used in combination with other agents for the treatment of sarcomas and a variety of carcinomas(Wilms tumor).

- **Mechanism of action:**

- **Intercalation into DNA strands**, also inhibits **topoisomerase II**-catalyzed breakage reunion→ **irreparable DNA strand breaks** → **free radicals** → single strand scission of DNA.

- The latter mechanism is **responsible** for the drug's **cardiac toxicity**.



Acute
Cardiotoxicity



Early Stage
Chronic
Cardiotoxicity



Late Stage
Chronic
Cardiotoxicity

High Cardiotoxicity Reversibility

Low

Adverse effects

- Irreversible **dose-related cardiotoxicity**.
- Thorax irradiation and Trastuzumab (monoclonal AB) increase cardiac toxicity.
- ****Iron chelator dexrazone** may ***protect against cardiac toxicity***, also liposomal encapsulated formulation cause less cardiotoxicity.
- The drug causes **reddish** discoloration of urine.
- ****Common adverse effects of cytotoxic drugs are also encountered (GIT, myelodepression and alopecia).**